(i) while loop

The syntax of while loop in Python has the following syntax:

Syntax:

while <condition>:
    statements block 1
[else:
    statements block2]

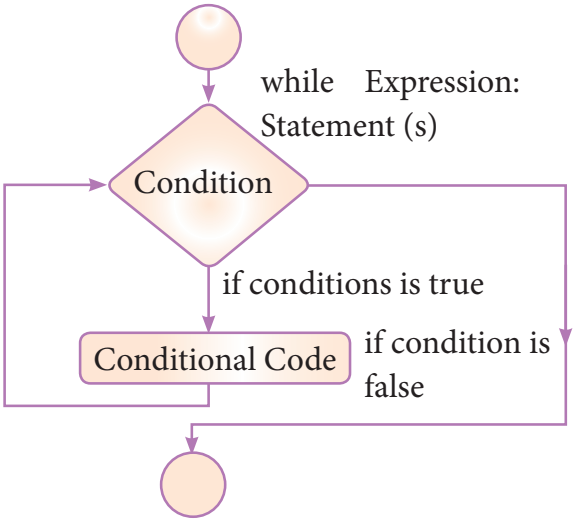

Fig 6.4 while loop execution

In the **while** loop, the condition is any valid Boolean expression returning True or False. The **else** part of while is optional part of **while**. The **statements block1** is kept executed till the condition is True. If the **else** part is written, it is executed when the condition is tested False. Recall **while** loop belongs to entry check loop type, that is it is not executed even once if the condition is tested False in the beginning.

Example 6.6: program to illustrate the use of while loop - to print all numbers from 10 to 15

```
i=10                    # intializing part of the control variable
while (i<=15):          # test condition
    print (i,end='\t')  # statements - block 1
    i=i+1               # Updation of the control variable
```

Output:

```
10    11    12    13    14    15
```